

Adiabatic Invariant in Mechanics

Definition

Adiabatic Invariant is defined as the physical quantity I that is *conserved* when some other parameters of the system λ varies with respect to time $\lambda = \lambda(t)$, and the changing rate $\dot{\lambda}$ is infinitesimal. In this case, the whole process can be considered as a simulated *adiabatic process* in thermodynamics.

Examples

Ex.1 Oscillator with Changing Spring Constant

We consider a spring oscillator with its spring constant changing with time in a slow rate. We assume that the constant mass is m , and the spring constant $k = k(t)$, to analyze the motion of the spring oscillator, we can write the total energy:

$$E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}k(t)x^2 \quad (1)$$

then differentiate the energy with respect to time:

$$\frac{dE}{dt} = m\dot{x}\ddot{x} + \frac{1}{2}\dot{k}x^2 + kx\dot{x} \quad (2)$$

By the second law of Newton, we obtain:

$$m\ddot{x} + kx = 0 \quad (3)$$

then substituting *Equ.3* into *Equ.2*, we obtain:

$$\frac{dE}{dt} = \frac{1}{2}\dot{k}x^2 \quad (4)$$

We then *average* the equation over one period T with respect to time, by the condition that k changes slowly, we can obtain during *the scale of one period*, the change can be considered as homogeneous, so the changing rate of the spring constant $\dot{k}$ is almost constant. Then the equation is:

$$\frac{d\langle E \rangle}{dt} = \frac{\dot{k}}{2}\langle x^2 \rangle \quad (5)$$

For a harmonic oscillator, we can prove that^[1] the average values of the potential energy and the kinetic energy are equal, so we have:

$$\left\langle \frac{kx^2}{2} \right\rangle = \left\langle \frac{m\dot{x}^2}{2} \right\rangle = \frac{1}{2}\langle E \rangle \quad (6)$$

So we can express $\langle x^2 \rangle$ in terms of $\langle E \rangle$ and k :

$$\langle x^2 \rangle = \frac{\langle E \rangle}{k} \quad (7)$$

then substituting *Equ.7* into *Equ.5*, we have:

$$\frac{d\langle E \rangle}{dt} = \frac{\langle E \rangle}{2} \frac{\dot{k}}{k} \quad (8)$$

So we can solve this ODE(ordinary differential equation) by separating variables and integrating on both sides, so the solution:

$$\ln \langle E \rangle = \frac{1}{2} \ln k + C \quad (9)$$

where C is a constant. Then we obtain:

$$\frac{\langle E \rangle}{\sqrt{k}} = C \quad (10)$$

So we determine a conserved quantity successfully, and we can rewrite the quantity in terms of $\langle E \rangle$ and $\omega = \sqrt{\frac{k}{m}}$:

$$\frac{\langle E \rangle}{\omega} = C' \quad (11)$$

where C' is also a constant, as the invariance of mass m . So the quantity

$$I = E/\omega$$

is the adiabatic invariant of the system.

Ex.2 Elastic Collisions of a Ball with a Wall and a Plate

• Also known as the simpler and original version of **Fermi accelerator**.

Consider the motion of a particle or ball between an elastic wall and an elastic heavy plate with a slow velocity u , the velocity of the ball is $\pm v$, there is no external force (including the gravity), and initially $u \ll v$. The motion of the ball is limited at the direction perpendicular to the wall and the plate (the wall is parallel to the plate), as shown in *Fig.1*, determine another conserved quantity during the motion.

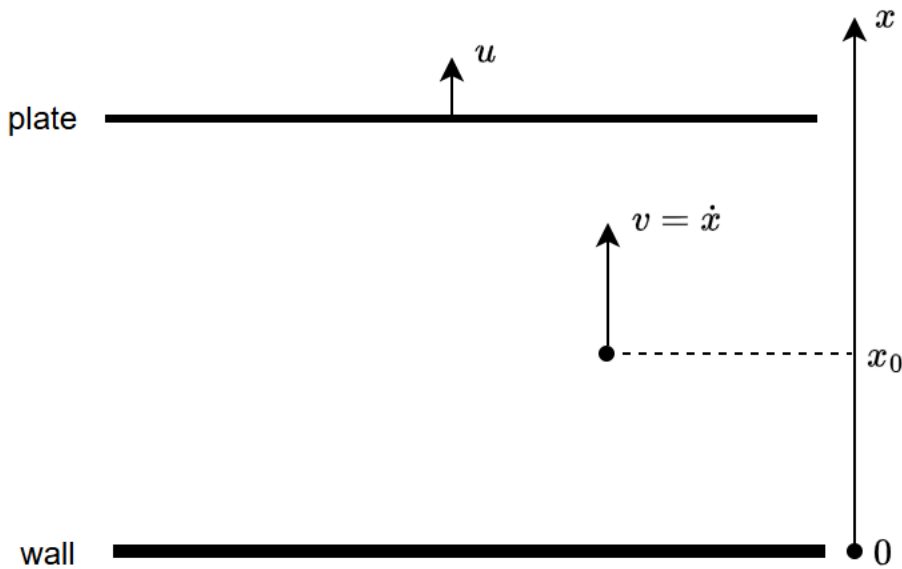


Fig.1: The motion and collision process

To solve this problem, we draw the *phase graph* of the system as shown in *Fig.2*, representing the relation between the velocity and the displacement as rectangles: at first, the ball moves with a constant speed $v = \dot{x}$, until it collides with the plate. After the collision, the velocity of the ball turns to the opposite direction, and have a loss of dv , but accordingly, the distance it needs to travel across increases by dx , so next time it collides with the plate, the graph of this period becomes another rectangle with less $\dot{x}$ and larger x .

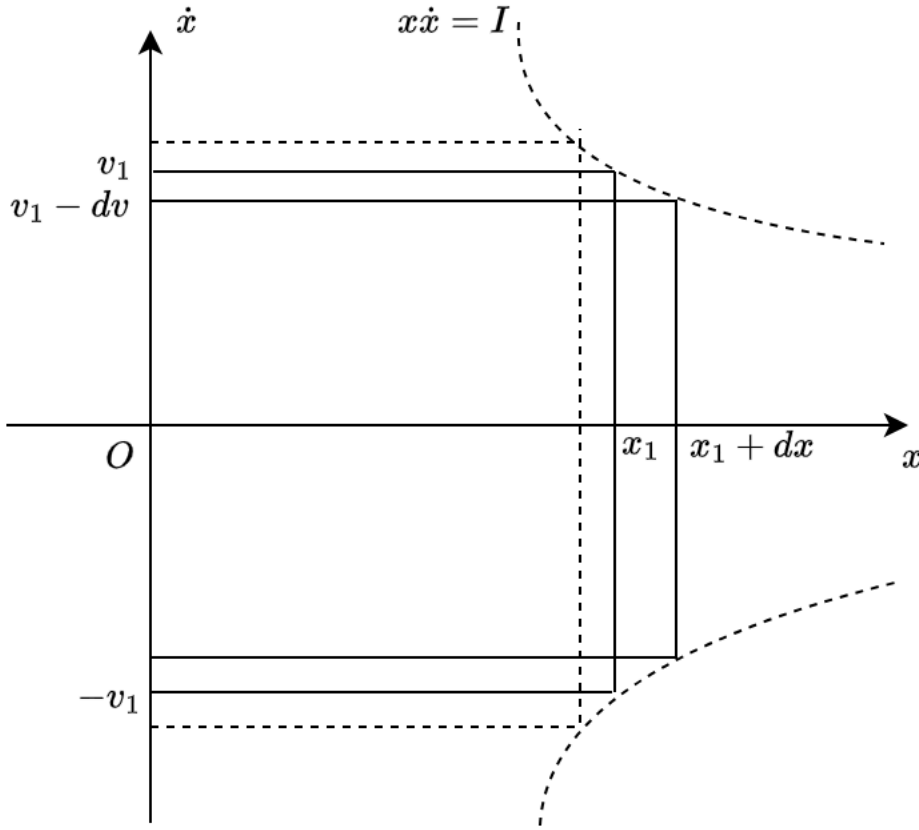


Fig.2: The graph of the system

Then we calculate the *subgraph*, or the area bounded by each rectangle, denoted by S . In one period, the velocity and the displacement are invariant, so we consider the motion including collisions, or namely, the motion represented by different rectangles, for instance, the velocity declines from v_1 to $v_1 - dv$, and the displacement increases from x_1 to $x_1 + dx$.

We denote the change of area of these two rectangles as ΔS , then we can obtain:

$$\Delta S = 2dx(v_1 - dv) - 2dvx_1 \quad (12)$$

By the conservation of momentum, we can obtain dv in terms of u ^[2]:

$$dv = 2u \quad (13)$$

As the speed of the motion in the same period is constant, we can obtain:

$$dx = u\Delta t \quad (14)$$

$$x_1 = v_1\Delta t \quad (15)$$

by the most basic formulas for constant-speed motion.

So substituting *Equ. 13*, *Equ. 14* and *Equ. 15* into *Equ. 12*, we obtain:

$$\Delta S = 2uv_1\Delta t - 2uv_1\Delta t - 2u^2\Delta t = 0 - 2u^2\Delta t \quad (16)$$

As $u \ll v$, we just drop the infinitesimal u^2 term, and obtain:

$$\Delta S = 0 \quad (17)$$

So we have found another conserved quantity in this motion:

$$I = x\dot{x}$$

More geometrically, it implies that the envelope of these rectangles is a hyperbola $x\dot{x} = I$, or all the vertex of the rectangle are located on a pair of hyperbola.

Theorem

Adiabatic Theorem of Landau

For an arbitrary dynamic system with Hamiltonian $H(q, p; \lambda)$, if the parameter λ varies with time $\lambda = \lambda(t)$ and the changing rate $\dot{\lambda} \rightarrow 0$, then we can define **action variable** as:

$$I = \frac{1}{2\pi} \oint p dq \quad (18)$$

It is approximately conserved with the changing of time^[3].

Tests in Previous Examples

Ex.1

The Hamiltonian is:

$$H = \frac{1}{2}k(t)x^2 + \frac{p^2}{2m} = E \quad (19)$$

So the generalized momentum and generalized coordinates is defined as:

$$\begin{cases} q = x \\ p = m\dot{x} \end{cases} \quad (20)$$

Then the action variable is:

$$I = \frac{1}{2\pi} \oint p dx \quad (21)$$

By *Equ.19*, we can solve for p as a function of x by defining $\omega = \sqrt{\frac{k}{m}}$:

$$p = \pm \sqrt{2mE - m^2\omega^2 x^2} \quad (22)$$

which represents an ellipse in $p - x$ phase graph. So we can calculate the integral simply by using the formula for the area of ellipses:

$$\oint p dx = \pi \cdot \frac{2E}{\omega} \quad (23)$$

So the action variable:

$$I = \frac{E}{\omega}$$

Ex.2

The Hamiltonian is:

$$H = \frac{p^2}{2m} \quad (24)$$

So it is easy to calculate the integral (as p and q are independent, the closed-loop integral only depends on the length x)

$$\oint p dq = 2xm\dot{x} \quad (25)$$

So the action variable is:

$$I' = \frac{m x \dot{x}}{\pi} = \frac{m}{\pi} I$$

1. The proof is attached as follows:

For a harmonic oscillator, its motion satisfies:

$$x(t) = A \sin\left(\sqrt{\frac{k}{m}}t + \varphi\right)$$

where $A = \sqrt{\frac{2E}{k}}$, φ is a constant.

Take the first order derivative of $x(t)$ with respect to t , then we obtain the velocity:

$$v(t) = A\sqrt{\frac{k}{m}} \cos\left(\sqrt{\frac{k}{m}}t + \varphi\right)$$

So the kinetic energy and the potential energy can be expressed as:

$$K(t) = \frac{1}{2}kA^2 \cos^2\left(\sqrt{\frac{k}{m}}t + \varphi\right)$$

$$U(t) = \frac{1}{2}kA^2 \sin^2\left(\sqrt{\frac{k}{m}}t + \varphi\right)$$

We take the average value with respect to time during one period T :

$$\langle K \rangle = \frac{\int_0^T K(t) dt}{T}$$

$$\langle U \rangle = \frac{\int_0^T U(t) dt}{T}$$

Then we obtain:

$$\langle K \rangle = \langle U \rangle = \frac{1}{4}kA^2$$



2. The derivation is attached as follows:

As the collision is elastic, we have the conservation of momentumn (relative to the plate) and energy:

$$mv + Mu = mv_1 + Mu_1$$

$$\frac{1}{2}mv^2 + \frac{1}{2}Mu^2 = \frac{1}{2}mv_1^2 + \frac{1}{2}Mu_1^2$$

When the plate is heavy ($m \ll M$), the value of the velocity is $v_1 = v - u$ in the reference frame of the plate, then back to the reference system of the ground, we have:

$$v_1 = v - 2u$$



3. The strict proof in mathematica is complex and advanced, so we do not discuss it here.